

RNA-seq Analysis of SARS-CoV-2 Infection

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2026-03-19

Introduction

RNA sequencing (RNA-seq) enables comprehensive analysis of gene expression across biological conditions (Wang et al., 2009). In this study, differential gene expression was performed to investigate the transcriptional response to SARS-CoV-2 using a publicly available RNA-seq dataset. The aim of the analysis was to identify significant genes that are differentially expressed and to determine their biological pathway in response to viral infection. It was hypothesised that SARS-CoV-2 infection induces differential expression of immune-related and cell cycle-associated genes.

Methods

Setup

```
library(apeglm)
library(ggplot2)
library(ggrepel)
library(pheatmap)
library(DESeq2)
library(enrichplot)
library(clusterProfiler)
library(org.Hs.eg.db)
library(AnnotationDbi)
```

Load Data

```
expr <- read.delim("GSE152418_p20047_Study1_RawCounts.txt", row.names = 1, check.names = FALSE)
colnames(expr) <- normalise_sample_ids(colnames(expr))
```

Metadata

```
samples <- colnames(expr)

if (file.exists(metadata_path)) {
  meta <- read.csv(metadata_path, row.names = 1, check.names = FALSE, stringsAsFactors = FALSE)
```

```

meta$sample <- normalise_sample_ids(meta$sample)
rownames(meta) <- meta$sample
meta <- meta[samples, , drop = FALSE]
} else {
  meta <- data.frame(sample = samples)
  meta$condition <- ifelse(grepl("nCoV|nCOV", samples), "Infected", "Control")
}

meta$condition <- factor(meta$condition, levels = c("Control", "Infected"))
rownames(meta) <- samples

if (any(is.na(meta$condition))) {
  stop("Some samples could not be assigned to a condition.")
}

sample_counts <- table(meta$condition)

```

Samples were grouped into infected and control conditions based on dataset annotations, with biological replicates present in each group. Where project metadata were available, these were used to confirm the sample labels before analysis. In the absence of explicit metadata annotation, sample conditions can be inferred from sample names, which may introduce classification bias.

Sample Summary

Table 1: Sample composition of the bulk RNA-seq dataset.

Condition	Number of samples
Control	17
Infected	17

Data Pre-processing

The count matrix was aligned to the sample metadata to ensure correct sample ordering. Lowly expressed genes were filtered out by retaining genes with a minimum total count across samples, reducing noise and improving statistical power for downstream analysis.

```

expr <- expr[, rownames(meta)]
genes_before_filter <- nrow(expr)
keep <- rowSums(expr) >= 10 # Filter out lowly expressed genes to reduce noise
expr <- expr[keep, ]
genes_after_filter <- nrow(expr)

```

After filtering, 37107 genes were retained from the original 60683 genes in the count matrix.

DESeq2 Analysis

Differential expression analysis was performed using DESeq2, a widely used method for RNA-seq analysis that models count data and controls false discovery rates (Love et al., 2014).

```

# Create DESeq2 dataset with condition as the main variable of interest

dds <- DESeqDataSetFromMatrix(
  countData = expr,
  colData = meta,
  design = ~ condition
)

dds <- DESeq(dds)

# Apply log2 fold change shrinkage to improve effect size estimation

res <- lfcShrink(dds, coef = "condition_Infected_vs_Control", type = "apeglm")
res <- res[!is.na(res$padj), ]

```

The design formula specified condition as the primary explanatory variable, enabling identification of genes associated with infection status. A log2 fold change threshold of 1 was applied to focus on biologically meaningful expression changes, while an adjusted p-value threshold of 0.05 was used to control for false discovery rate.

DESeq2 is widely used for RNA-seq analysis due to its robust modelling of count-based data and control of false discovery rates.

Visualisation

PCA

```

vsd <- vst(dds, blind = FALSE)
pca_plot <- plotPCA(vsd, intgroup = "condition")
pca_plot +
  labs(title = "PCA of Gene Expression")

```

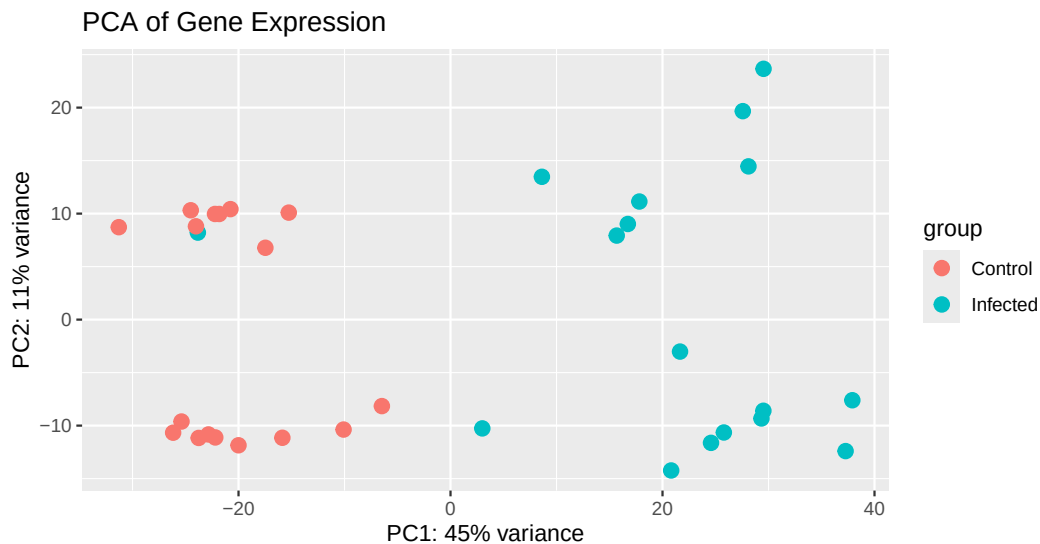


Figure 1: Principal Component Analysis (PCA) showing separation between infected and control samples, indicating that infection status is a major source of variation in gene expression.

Volcano Plot

```
res_df <- as.data.frame(res)

res_df$symbol <- mapIds(
  org.Hs.eg.db,
  keys = gsub("\\\\.*", "", rownames(res_df)),
  column = "SYMBOL",
  keytype = "ENSEMBL",
  multiVals = "first"
)
res_df$symbol[is.na(res_df$symbol)] <- gsub("\\\\.*", "", rownames(res_df))[is.na(res_df$symbol)]

res_df$group <- "NS"
res_df$group[res_df$padj < 0.05 & res_df$log2FoldChange > 1] <- "Up"
res_df$group[res_df$padj < 0.05 & res_df$log2FoldChange < -1] <- "Down"

top_genes <- head(rownames(res_df[order(res_df$padj), ]), 10)
label_data <- res_df[rownames(res_df) %in% top_genes, ]

ggplot(res_df, aes(x = log2FoldChange, y = -log10(padj), colour = group)) +
  geom_point(alpha = 0.5, size = 1) +

  geom_text_repel(
```

```

data = label_data,
aes(label = symbol),
size = 3,
max.overlaps = Inf,
box.padding = 0.6,
point.padding = 0.4,
segment.color = "black",
segment.size = 0.3,
force = 2,
show.legend = FALSE
) +

scale_colour_manual(values = c("Down" = "navy", "Up" = "firebrick3", "NS" = "gray")) +
geom_vline(xintercept = c(-1, 1), linetype = "dashed") +
geom_hline(yintercept = -log10(0.05), linetype = "dashed") +
labs(title = "Volcano Plot of Differential Expression") +
theme_minimal()

```

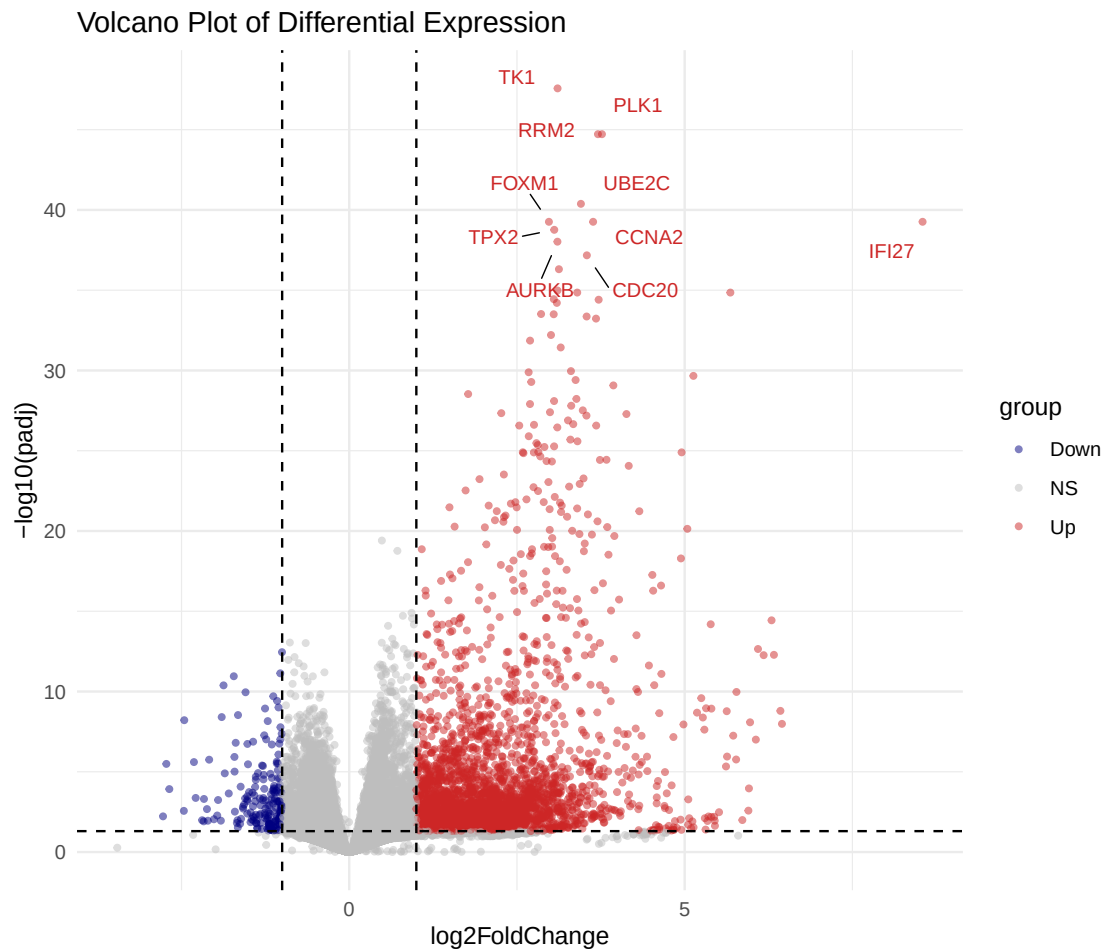


Figure 2: Volcano plot of differential gene expression showing $\log_2$ fold change versus statistical significance. Thresholds of $|\log_2\text{FC}| > 1$ and adjusted p-value < 0.05 are indicated by dashed lines. Red points represent upregulated genes, blue points represent downregulated genes, and grey points are not significant.

MA Plot

```
res_df <- as.data.frame(res)

res_df$significant <- "NS"
res_df$significant[res_df$padj < 0.05 & abs(res_df$log2FoldChange) > 1] <- "Significant"

ggplot(res_df, aes(x = baseMean, y = log2FoldChange, colour = significant)) +
  geom_point(alpha = 0.3, size = 0.9) +
  scale_x_log10() +
  scale_colour_manual(values = c("NS" = "grey", "Significant" = "red")) +
  geom_hline(yintercept = 0, colour = "black") +
  geom_hline(yintercept = c(-1, 1), linetype = "dashed") +
  labs(
    title = "MA Plot of Differential Expression",
    x = "Mean of normalized counts",
    y = "Log2 fold change",
    colour = "Significance"
  ) +
  theme_minimal()
```

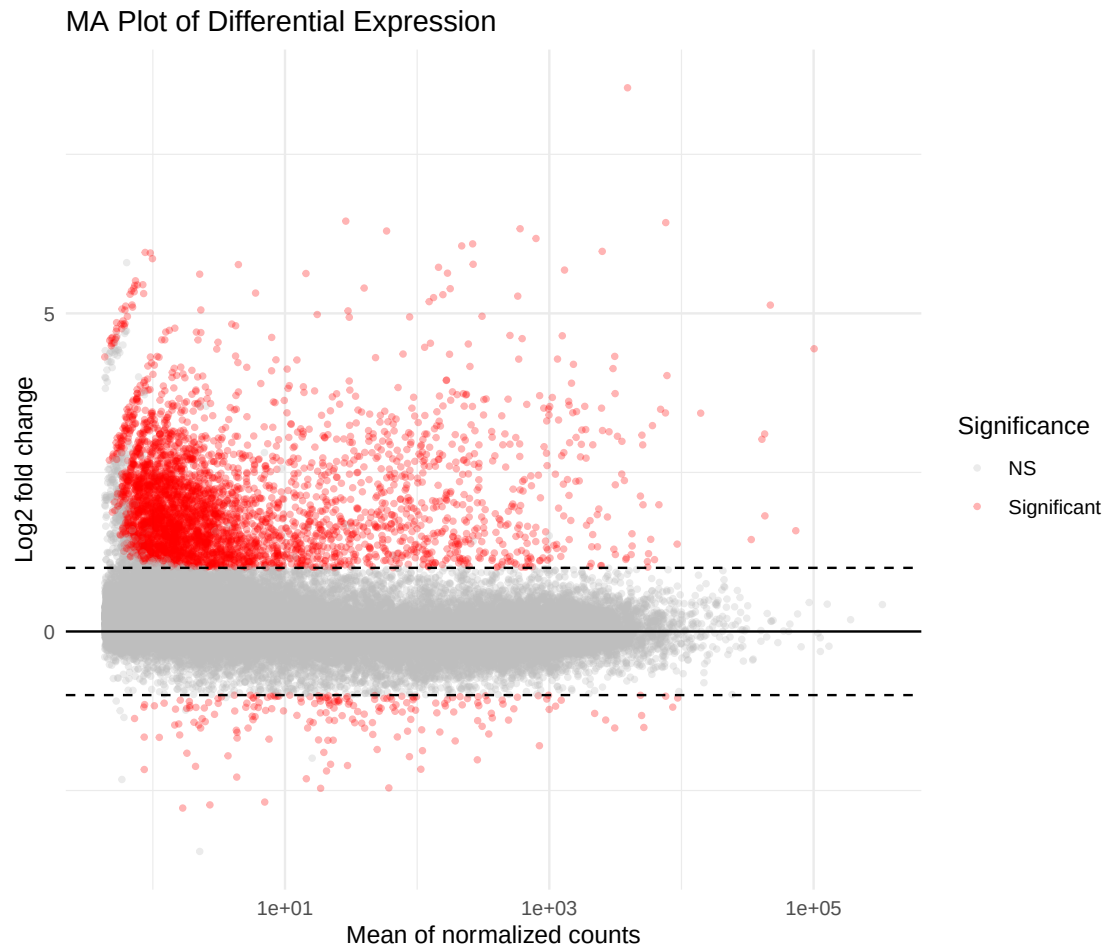


Figure 3: MA plot showing the relationship between mean expression and log2 fold change, demonstrating that differential expression occurs across a wide range of expression levels.

Heat-map

```
top_genes <- rownames(res)[order(res$padj)][1:20]

# Convert ENSEMBL -> SYMBOL
gene_symbols <- mapIds(
  org.Hs.eg.db,
  keys = gsub("\\\\.*", "", top_genes),
  column = "SYMBOL",
  keytype = "ENSEMBL",
  multiVals = "first"
)
gene_symbols[is.na(gene_symbols)] <- gsub("\\\\.*", "", top_genes)[is.na(gene_symbols)]

heatmap_data <- assay(vsd)[top_genes, ]
rownames(heatmap_data) <- gene_symbols

annotation_col <- data.frame(condition = meta$condition)
```

```

rownames(annotation_col) <- rownames(meta)

pheatmap(
  heatmap_data,
  scale = "row",
  color = colorRampPalette(c("navy", "white", "firebrick3"))(50),
  annotation_col = annotation_col,
  cluster_rows = TRUE,
  cluster_cols = TRUE,
  show_colnames = FALSE,
  fontsize_row = 8
)

```

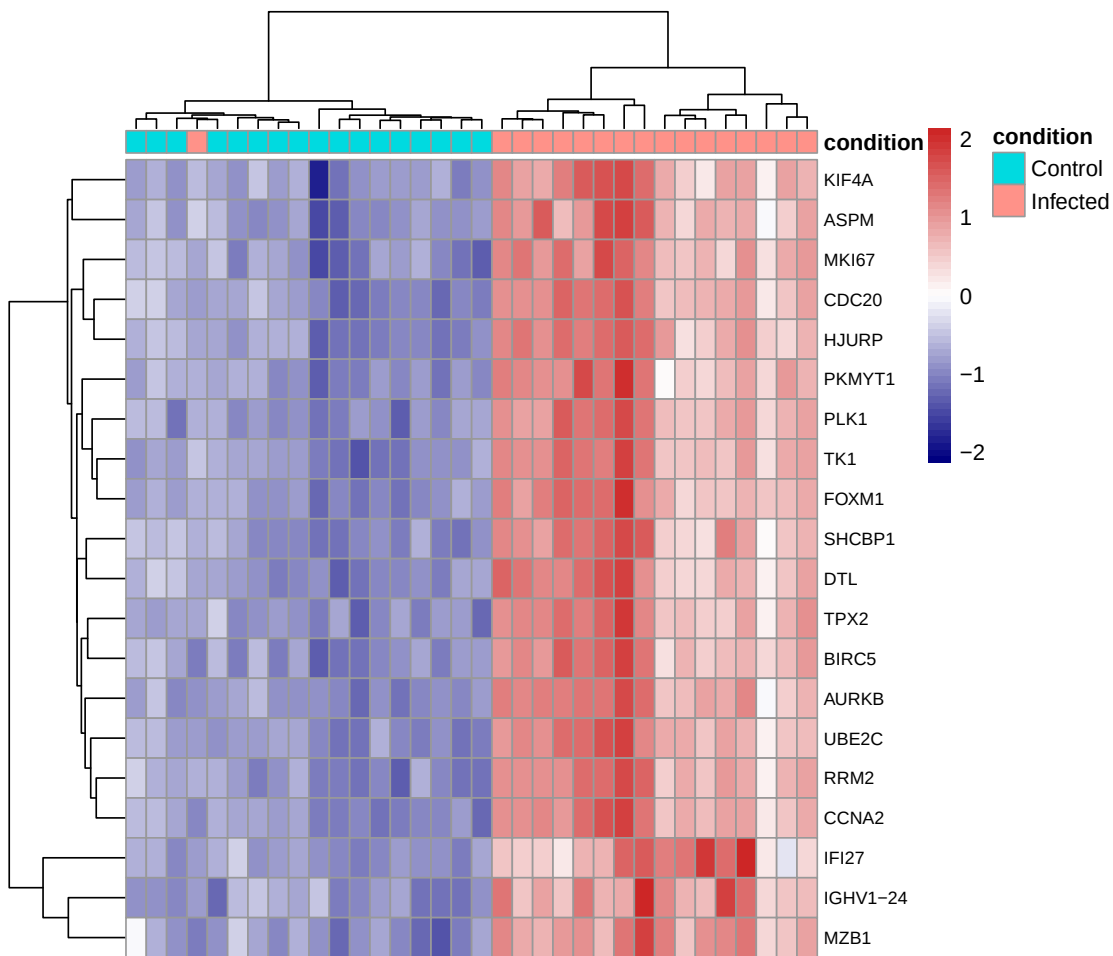


Figure 4: Heatmap of the top 20 differentially expressed genes showing distinct clustering of infected and control samples, indicating consistent expression patterns within each condition.

Sample Distance Heat-map

```

sampleDists <- dist(t(assay(vsd)))

```

```
pheatmap(
  as.matrix(sampleDists),
  annotation_col = annotation_col,
  color = colorRampPalette(c("navy", "white", "firebrick3"))(50),
  show_colnames = FALSE,
  show_rownames = FALSE,
)
```

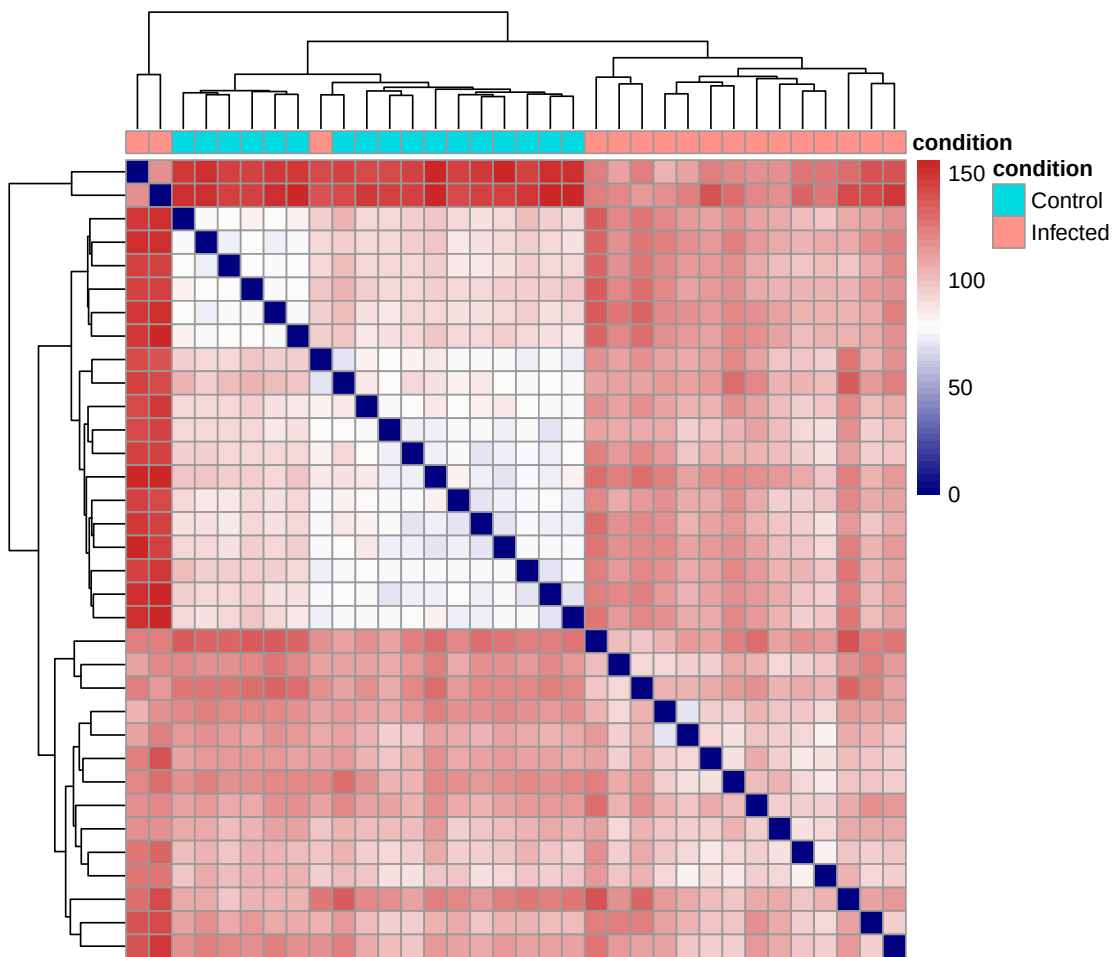


Figure 5: Sample-to-sample distance heatmap demonstrating clustering based on global gene expression profiles, with infected samples grouping separately from controls.

GSEA (GO)

Functional enrichment analysis was conducted using the clusterProfiler package (Yu et al., 2012).

Gene set enrichment analysis was performed to identify coordinated biological pathways associated with SARS-CoV-2 infection (Subramanian et al., 2005).

```
genes <- gsub("\\\\.\\.", "", rownames(res))
gene_list <- res$log2FoldChange
```

```

names(gene_list) <- genes
gene_list <- gene_list[!is.na(gene_list)]

# Convert ENSEMBL IDs to ENTREZ IDs required for GSEA

gene_df <- bitr(
  names(gene_list),
  fromType = "ENSEMBL",
  toType = "ENTREZID",
  OrgDb = org.Hs.eg.db
)

gene_df <- gene_df[!is.na(gene_df$ENTREZID), ]
gene_df <- gene_df[!duplicated(gene_df$ENTREZID), ]

gene_list <- gene_list[gene_df$ENSEMBL]
names(gene_list) <- gene_df$ENTREZID
gene_list <- sort(gene_list, decreasing = TRUE) # Rank genes by log2 fold change for GSEA input

ego_gsea <- gseGO(
  geneList = gene_list,
  OrgDb = org.Hs.eg.db,
  keyType = "ENTREZID",
  ont = "BP",
  verbose = FALSE
)

top_pathways <- head(ego_gsea@result$Description, 2)
top_pathway_text <- paste(top_pathways, collapse = " and ")

dotplot(ego_gsea, showCategory = 8)

```

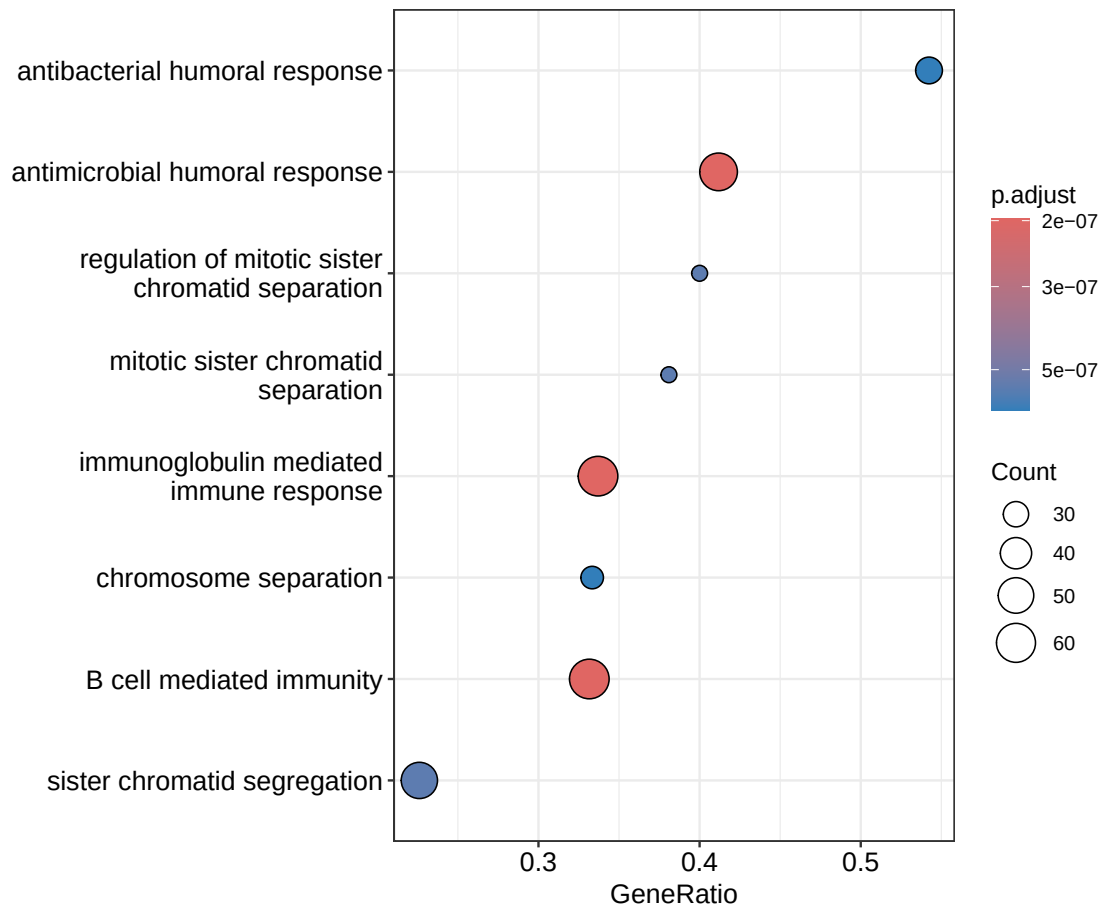


Figure 6: Gene Set Enrichment Analysis (GSEA) dotplot showing significant enrichment of immune-related and cell cycle-associated biological processes.

Enrichment curve (top pathway)

```
gseaplot2(ego_gsea, geneSetID = 1)
```

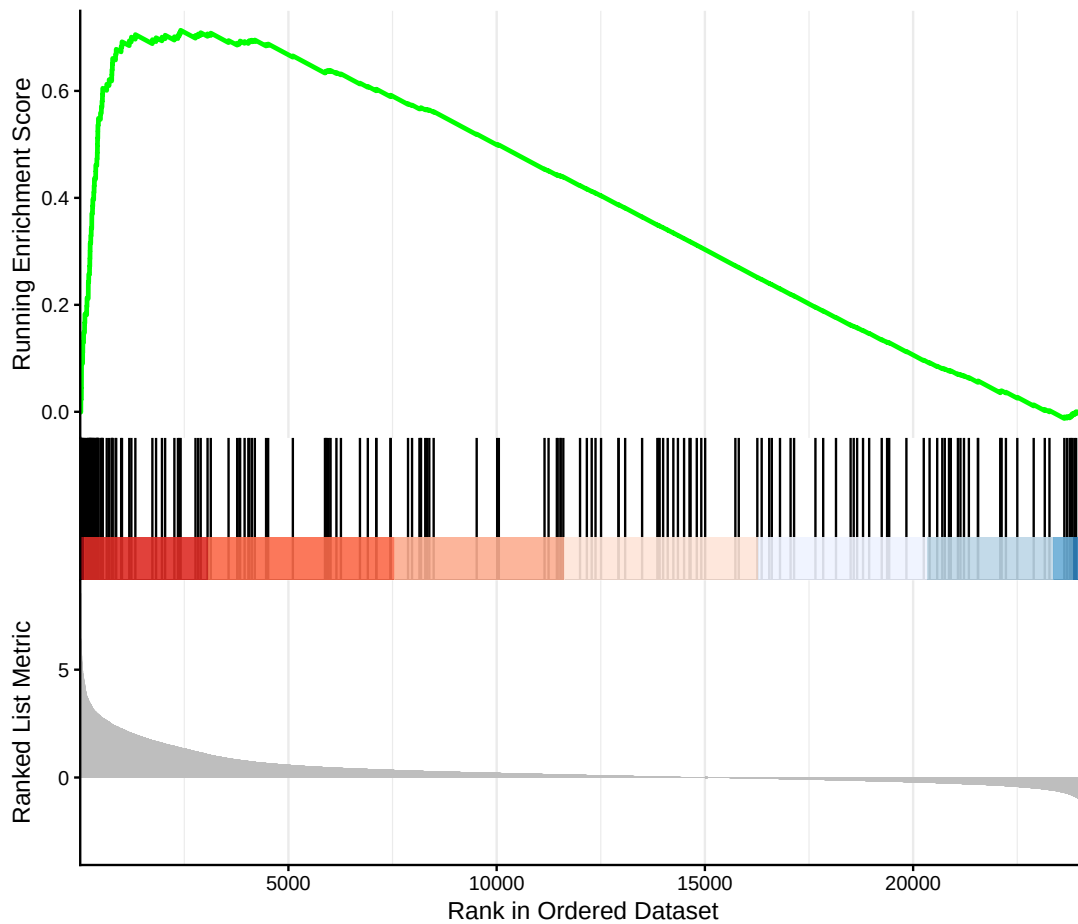


Figure 7: GSEA enrichment plot for the top enriched pathway, illustrating the distribution of pathway-associated genes across the ranked gene list and their contribution to the enrichment score.

Results

This dataset comprised 17 infected samples and 17 control samples, allowing direct comparison of transcriptional responses associated with SARS-CoV-2 infection.

Differentially Expressed Genes

```
sig <- res[res$padj < 0.05 & abs(res$log2FoldChange) > 1, ]
n_sig <- nrow(sig)
n_up <- sum(sig$log2FoldChange > 0)
n_down <- sum(sig$log2FoldChange < 0)
```

Differential Expression Summary

Table 2: Differential expression summary using the report threshold of adjusted p-value < 0.05 and $|\log_2 \text{fold change}| > 1$.

Metric	Count
Significant DEGs	3879
Up-regulated genes	3688
Down-regulated genes	191

Top Differentially Expressed Genes

Table 3: Top five differentially expressed genes ranked by adjusted p-value.

	Gene	Log2 fold change	Adjusted p-value
ENSG00000167900	TK1	3.106	0
ENSG00000171848	RRM2	3.707	0
ENSG00000166851	PLK1	3.767	0
ENSG00000175063	UBE2C	3.453	0
ENSG00000145386	CCNA2	3.635	0

Enriched Biological Pathways

Table 4: Top enriched biological pathways identified by GSEA.

	Pathway	NES	Adjusted p-value
GO:0016064	immunoglobulin mediated immune response	1.657	0
GO:0019724	B cell mediated immunity	1.642	0
GO:0019730	antimicrobial humoral response	1.635	0
GO:0010965	regulation of mitotic sister chromatid separation	1.772	0
GO:0051306	mitotic sister chromatid separation	1.764	0
GO:0000819	sister chromatid segregation	1.486	0

	Pathway	NES	Adjusted p-value
The results from the differential genes revealed that SARS-CoV-2 infection results in a marked transcriptional response, where 3688 genes were found to be up-regulated, whereas 191 were found to be down-regulated under the stricter significance threshold used in this report (adjusted p-value < 0.05 and	log2 fold change	> 1).	

Thus, infection status was a major independent source of variation within the dataset, as revealed by PCA, which clearly distinguished between uninfected and infected samples (Figure 1).

Volcano and MA plots show there are a large proportion of up-regulated genes among the significantly differentially expressed genes, suggesting widespread transcriptional activation associated with infection. The volcano plot (Figure 2) shows a strong shift in transcription in favour of upregulation, signalling activated infection-responsive biological pathways. The MA plot also shows that differential expression occurs across a wide range of gene expression levels.

Most importantly, TK1, RRM2, PLK1, UBE2C, CCNA2 were linked to DNA replication and cell cycle progression. The activation of the genes points to a cell cycle-associated transcriptional program, and this could also be an indication of immune cell expansion, or a dysregulation of host cells. The heat map of the major differentially expressed genes (Figure 4) shows a grouping together of the infected and controls based on distinct patterns of gene expression under the two conditions. This is further supported by the sample-to-sample distance analysis (Figure 5), with samples from infected populations clustered together and distinguished from controls.

The Gene Set Enrichment Analysis (GSEA) results (Figure 6) indicated significant enrichment of pathways such as immunoglobulin mediated immune response and B cell mediated immunity. The cell-cycle pathways are enriched alongside immune-related pathways, supporting a mixed host-response signature in the infected samples.

Overall, the increase in immune pathways and upregulation of genes associated with the cell cycle may indicate activation of immune processes. This observation may also suggest an increase in proliferating cells or changes in cell composition.

Discussion

The immune pathways were found to be more active as part of the host response to the virus. SARS-CoV-2 infection activates both innate and adaptive immune responses, including B cell activation and antibody production. Some of the enriched pathways include immunoglobulin-mediated and humoral immune processes, which point to the transcriptional effect of adaptive immune activity.

Infection-related immune responses, notably by lymphocytes, may reflect increased expression of cell cycle-associated genes, potentially indicating cellular proliferation or shifts in cell population composition, reflected by expression of cell cycle genes including PLK1, CCNA2, and RRM2. This may represent virus-induced cellular stress responses or compensatory mechanisms initiated by viral replication and host cell damage.

The heat-map and sample distance analysis provided further support for these observations. There was also a consistent expression pattern among the top DEGs across samples. This shows that the transcriptional reaction to infection is strong and reproducible.

The results are consistent with those of other viral infections, where activation of immune responses and cellular activation are key features of the host response. In the case of the strong cell cycle signal, further investigation would be necessary to determine whether its relevance reflects expansion of immune cells or dysregulation of host cellular machinery as a consequence of viral infection.

Critical Evaluation

Despite the effectiveness of this analysis, there are several limitations to be taken into account. The use of bulk RNA-seq limits the resolution of cell-type-specific transcriptional responses, making it difficult to determine whether observed changes arise from shifts in cell composition or true transcriptional regulation within individual cell types. This distinction is critical for viral-induced immune activation because there is a chance that the observed effect is due to immune cell infiltration rather than a transcriptional regulation event.

Second, the analysis models infection status as the primary explanatory variable. However, sequencing depth, batch effects, and sample heterogeneity may also contribute to observed variation, and other confounding factors may affect the findings and their interpretation. Thus, technical confounds may influence the results even though DESeq2 accounts for some sources of technical variation.

Other than that, an aspect that can affect the identified genes includes the thresholds chosen for significance (adjusted p-value < 0.05 and log2 fold change > 1). Differential enrichment pathways may also be observed following the use of different thresholds.

Also, the use of gene-level summarisation may hide issues with isoform-specific regulation, which may be relevant in viral infection contexts where alternative splicing can occur.

In the end, functional interpretation is left in the hands of annotation databases such as Gene Ontology that can be biased or incomplete to well-characterised genes. It is, therefore, appropriate always to interpret enrichment results cautiously and validate them experimentally when possible.

Future Steps

The results need to be further refined and validated by future studies. This would be a particularly valuable area for follow-up because scRNA-seq can resolve cell-type-specific transcriptional responses. The enrichment of immune- and cell cycle-associated pathways points to a biologically meaningful host response, providing

further evidence that adaptive immune activation and proliferative programs may both contribute to the SARS-CoV-2 transcriptional signature.

Also, if we integrate proteomic or metabolomic data it can also give us an indication of how transcriptional changes manifesting in functional cell responses. Integrating different omics data would enhance biological interpretation of the findings and identify key regulatory mechanisms during infection.

It is also important to validate experimentally the identified differentially expressed genes and pathways. For instance, quantitative PCR (qPCR) or flow cytometry tools could be used to validate gene expression changes and examine immune cell activation in independent samples.

From a computational perspective, alternative statistical models or machine learning approaches could also be applied to identify infection biomarkers and predictive signatures. For instance, the gene expression profiles of control and infected samples could be modelled to discriminate between the two sample types.

Another analysis that would be helpful is longitudinal or time-course data analyses that understand the timing of the host response to SARS-CoV-2 infection. As a result, identification of early and late response and characterization of disease progression can be done.

Supplementary Outputs

Updated DEG and enrichment result tables were exported to the **Data** folder to support downstream inspection and reproducibility.

Conclusion

In conclusion, this study demonstrates extensive differential gene expression associated with SARS-CoV-2 infection. There is also enhanced expression of cell cycle-related genes, with enrichment analysis showing activation of immune-associated pathways in infected samples.

The results show that host immune activation and increased cell cycle activity occur simultaneously in response to viral infection.

Collectively, these findings suggest coordinated crosstalk between immune system activation and cell cycle control following viral infection. This reveals biologically meaningful circuitry that warrants further investigation in future studies.

References

- Love, M. I., Huber, W., and Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology*, 15(12), 550.
- Wang, Z., Gerstein, M., and Snyder, M. (2009). RNA-Seq: a revolutionary tool for transcriptomics. *Nature Reviews Genetics*, 10(1), 57-63.
- Subramanian, A., Tamayo, P., Mootha, V. K., et al. (2005). Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences*, 102(43), 15545-15550.
- Yu, G., Wang, L.-G., Han, Y., and He, Q.-Y. (2012). clusterProfiler: An R package for comparing biological themes among gene clusters. *OMICS*, 16(5), 284-287.

Reproducibility

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_Ireland.utf8  LC_CTYPE=English_Ireland.utf8
## [3] LC_MONETARY=English_Ireland.utf8 LC_NUMERIC=C
## [5] LC_TIME=English_Ireland.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats4      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] org.Hs.eg.db_3.22.0      AnnotationDbi_1.72.0
## [3] clusterProfiler_4.18.4  enrichplot_1.30.5
## [5] DESeq2_1.50.2            SummarizedExperiment_1.40.0
## [7] Biobase_2.70.0          MatrixGenerics_1.22.0
## [9] matrixStats_1.5.0       GenomicRanges_1.62.1
## [11] Seqinfo_1.0.0           IRanges_2.44.0
## [13] S4Vectors_0.48.0       BiocGenerics_0.56.0
## [15] generics_0.1.4         pheatmap_1.0.13
## [17] ggrepel_0.9.8          ggplot2_4.0.2
## [19] apeglm_1.32.0
##
## loaded via a namespace (and not attached):
## [1] RColorBrewer_1.1-3      rstudioapi_0.18.0      jsonlite_2.0.0
## [4] tidyr_0.0.6            magrittr_2.0.4         ggtangle_0.1.1
## [7] farver_2.1.2           rmarkdown_2.30        fs_1.6.7
## [10] vctrs_0.7.1            memoise_2.0.1          ggtree_4.0.5
## [13] htmltools_0.5.9        S4Arrays_1.10.1        SparseArray_1.10.9
## [16] gridGraphics_0.5-1     htmlwidgets_1.6.4      plyr_1.8.9
## [19] cachem_1.1.0           igraph_2.2.2           lifecycle_1.0.5
## [22] pkgconfig_2.0.3        Matrix_1.7-4           R6_2.6.1
## [25] fastmap_1.2.0          gson_0.1.0             digest_0.6.39
## [28] numDeriv_2016.8-1.1    aplot_0.2.9            ggnewscale_0.5.2
## [31] patchwork_1.3.2        RSQLite_2.4.6          labeling_0.4.3
## [34] httr_1.4.8            polyclip_1.10-7        abind_1.4-8
## [37] compiler_4.5.2         bit64_4.6.0-1          fontquiver_0.2.1
## [40] withr_3.0.2            S7_0.2.1              BiocParallel_1.44.0
## [43] DBI_1.3.0              ggforce_0.5.0          R.utils_2.13.0
## [46] MASS_7.3-65            rappdirs_0.3.4         DelayedArray_0.36.0
## [49] tools_4.5.2           otel_0.2.0            ape_5.8-1
## [52] scatterpie_0.2.6       R.oo_1.27.1           glue_1.8.0
## [55] nlme_3.1-168           GOSemSim_2.36.0        grid_4.5.2
## [58] cluster_2.1.8.1        reshape2_1.4.5        snow_0.4-4
```

## [61] fgsea_1.36.2	gtable_0.3.6	R.methodsS3_1.8.2
## [64] tidyr_1.3.2	data.table_1.18.2.1	XVector_0.50.0
## [67] pillar_1.11.1	stringr_1.6.0	yulab.utils_0.2.4
## [70] emdbook_1.3.14	splines_4.5.2	dplyr_1.2.0
## [73] tweenr_2.0.3	treeio_1.34.0	lattice_0.22-7
## [76] bit_4.6.0	tidyselect_1.2.1	fontLiberation_0.1.0
## [79] GO.db_3.22.0	locfit_1.5-9.12	Biostrings_2.78.0
## [82] knitr_1.51	fontBitstreamVera_0.1.1	xfun_0.56
## [85] stringi_1.8.7	lazyeval_0.2.2	ggfun_0.2.0
## [88] yaml_2.3.12	evaluate_1.0.5	codetools_0.2-20
## [91] bbmle_1.0.25.1	gdtools_0.5.0	tibble_3.3.1
## [94] qvalue_2.42.0	ggplotify_0.1.3	cli_3.6.5
## [97] systemfonts_1.3.2	Rcpp_1.1.1	coda_0.19-4.1
## [100] png_0.1-9	bdsmatrix_1.3-7	parallel_4.5.2
## [103] blob_1.3.0	DOSE_4.4.0	mvtnorm_1.3-6
## [106] tidytree_0.4.7	ggiraph_0.9.6	scales_1.4.0
## [109] purrr_1.2.1	crayon_1.5.3	rlang_1.1.7
## [112] cowplot_1.2.0	fastmatch_1.1-8	KEGGREST_1.50.0